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Development of wound therapy in nursing care of infants by using injectable gelatin-cellulose composite hydrogel incorporated with silver nanoparticles

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A B S T R A C T

Nano silver material has been regarded as a kind of promising candidate for biomedical application due to its low cytotoxicity and striking antibacterial activity. In this study, an injectable silver-gelatin-cellulose ternary hydrogel dressing was prepared by the introduction of gelatin and aminated silver nanoparticles into the carboxylated cellulose nanofibers for efficient wound therapy in infant nursing. The produced silver nanoparticles and composite hydrogel were characterized by using transmission electron microscopy (TEM), scanning electron microscopy (SEM), Thermogravimetric analysis (TGA), and X-ray diffraction (XRD) techniques. The as-synthesized silver-cellulose-gelatin ternary hydrogel demonstrated high porosity and satisfactory antibacterial activity. In vitro cytocompatibility measurement revealed the very low toxic nature of such composite hydrogel dressing. Additionally, the in vivo assessment in mice confirmed that the silver-cellulose-gelatin ternary dressing possessed improved wound healing ability. Hence, this study strongly supports the possibility of using this novel silver-gelatin-cellulose ternary dressing for efficient cutaneous wound healing in nursing care of infants.

1. Introduction

As a natural barrier organ, skin plays an important role in protecting infant body, particularly the prevention of external invasion by microorganisms [1,2]. Injured skin can't ensure its barrier function, in which harmful microbes could invade easily to bring a series of damages. In skin wound healing administration, bacterial infection is a main challenge that can hinder the wound recuperating, induce exudates development, and contribute to inappropriate collagen formation [3,4]. In order to control infections or remove excessive exudates, the systemic antibiotics are not primarily recommended since they may lead to sensitizing side effects and bacterial resistant [5,6]. For this concern, exploration and development of efficient wound dressing with outstanding antibacterial activity is of great significance for wound healing in clinical nursing care of infants. An ideal wound dressing needs to keep moist at the injury interface and have desirable biocompatibility and gas permeability [7]. Accordingly, various dressings have been designed with porous structures to meet the demands of wound exudate absorption, controlled water vapor evaporation, and gas permeability [8–11].

Natural biopolymers such as chitosan, cellulose, collagen, and gelatin have been widely investigated for the preparation of wound dressings because of their hydrophilicity and non-cytotoxicity [12–14]. Especially, as one of the most abundant natural polymers, cellulose and its derivatives demonstrate excellent biocompatibility through variety of in vitro and in vivo measurements, thus being considered as promising candidates for biological and medical applications [15,16]. However, the traditional cellulose gauze-based dressings show limited healing capability according to the clinical practices, leading to theirs low efficiency for preventing bacteria growth and absorbing the exudates [17,18]. Therefore, development of new type of cellulose-based wound dressings presents the possibility to effectively facilitate wound healing.

Herein, we report the fabrication of antibacterial silver-gelatin-cellulose (CNF/G/Ag) nanofiber ternary hydrogel toward efficient wound-healing in nursing care of infants. In this composite, the hydrogen bonding interactions between cellulose and gelatin promote the formation of three-dimensional (3D) porous cross-linking scaffold, and the Ag nanoparticles show strong antimicrobial activity, the synergistic effect enables such ternary wound dressing to deliver enhanced wound healing effect.

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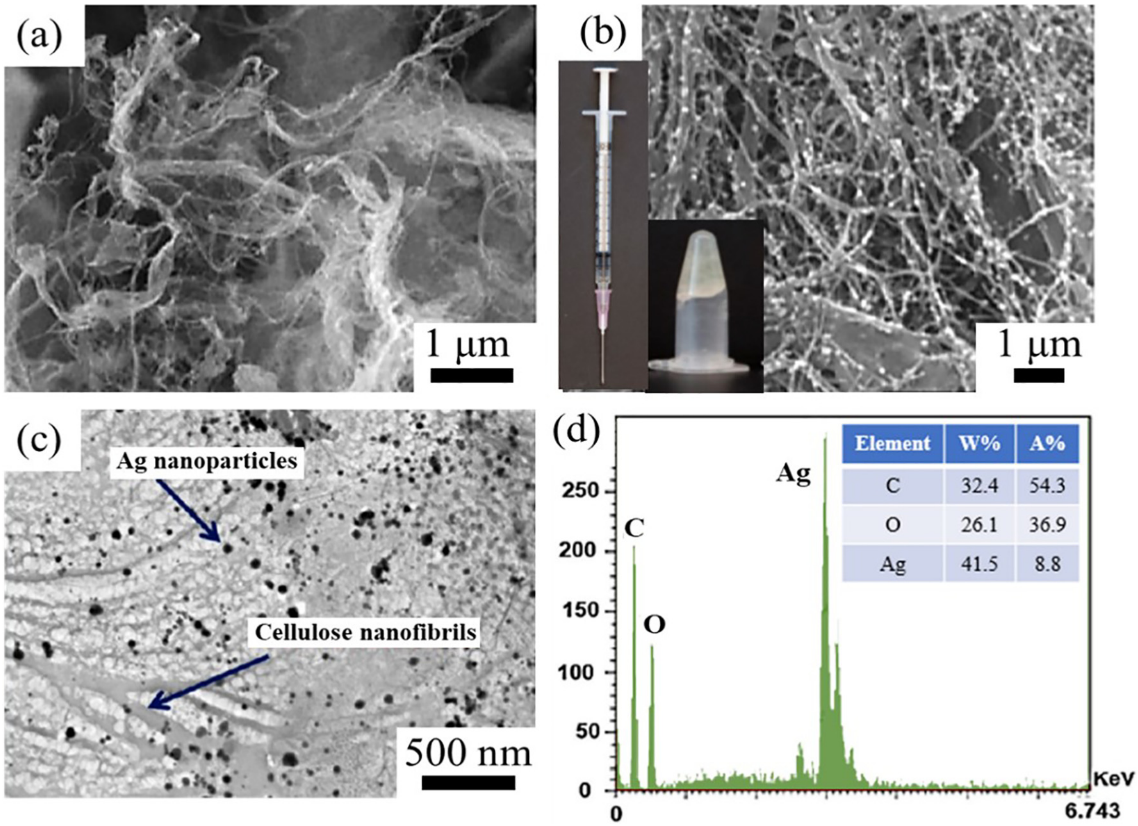

Fig. 1. (a) SEM image of CNF/G binary hydrogel. SEM (b) and TEM (c) images of CNF/G/Ag ternary hydrogel, the inset is its optical image. (d) EDS spectrum of CNF/G/Ag ternary hydrogel.

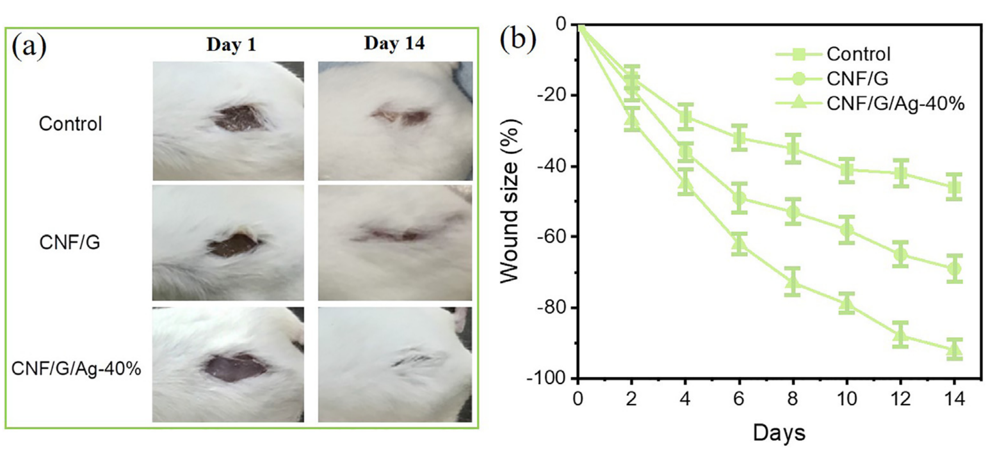

Fig. 2. (a) In vivo study of blank, CNF/G, and CNF/G/Ag nanocomposite as wound dressing, and the evaluation of wound size (b). n = 5 for all groups, values are mean ± SE.

2. Experimental section

Aminated Ag nanoparticles (NPs) were synthesized according to a previously reported procedure [19]. Briefly, a certain amount of $AgNO_3$ and polyvinyl pyrrolidone (PVP) was dispersed in 200 mL EG solution under stirring, the above uniform solution was heated at 120 °C for 1 h. The Ag NPs were collected after a followed reaction in 0.5 mol/L NaOH aqueous solution at 70 °C for 12 h. The products were collected and washed with distilled water and ethanol, respectively. After the redispersion in water/ethanol (1/9 v/v) solution, the aminated Ag NPs were obtained via 3-aminopropyltriethoxysilane (APTES) treatment at another 70 °C for 12 h, which were then collected, washed, and dried. For injectable silver-cellulose-gelatin ternary hydrogel, 200 mg of carboxylated CNF was firstly dissolved in 14 mL distilled water under stirring. Next, aminated Ag NPs and gelatin solution (80 mg/mL) were added into the above solution by ultrasonic for 30 min. The entire mixture was then kept at room temperature under vigorous stirring for 3 h to obtain cross-linked hydrogels.

3. Results and discussion

Fig. S1a shows the TEM image of as-prepared aminated Ag nanoparticles (NPs), which have the uniform diameter mainly

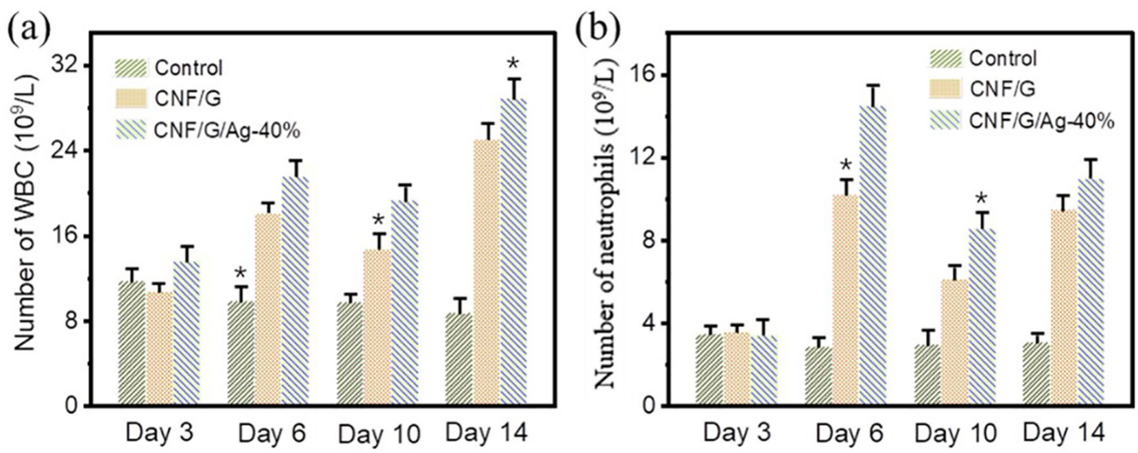

Fig. 3. In vivo assessment of the numbers of white blood cells (a) and neutrophils (b). n = 5 for all groups, values are mean ± SE. Statistical comparisons were conducted using one-way ANOVA, followed by Tukey test. $^*p < 0.05$ vs. control group.

around 30–60 nm (Fig. S1b). Carboxylated CNF with negative-charged surface is able to interact with the aminated Ag NPs through the electrostatic forces. Additionally, van der Waals interactions and hydrogen bonds can also promote the formation of CNF/G/Ag ternary hydrogel with homogeneous interpenetrating polymeric network structure. The chemical interactions between components in such composite dressing can sustain the appropriately release of the Ag NPs with prolonged activities. As we can see from the TGA curves, (Fig. S2), we fabricated various hydrogels with different mass content of Ag NPs. Fig. S3 depicts the XRD profiles of CNF/G and CNF/G/Ag composites. Three characteristic peaks located at the 2θ degrees of 37.9°, 44.0°, and 64.2° are ascribed to the (111), (200), and (220) crystallographic planes of face-centered cubic Ag (JCPDS file no. 04-0783) [20].

To present a better understanding of CNF/G/Ag ternary hydrogel, the microscopic morphology and elemental information have been characterized. Fig. 1a offers the SEM image of the CNF/G sample, which is composed of plenty of nanofibrils with about 50–200 nm in diameter and several micrometers in length. The aminated Ag NPs were uniformly anchored by the polymer matrix, as revealed by SEM image (Fig. 1b) and TEM image (Fig. 1c). The inset is the optical photo of the injectable CNF/G/Ag hydrogel, suggesting its great potential to use as efficient dressing for wound healing. The EDS spectra for CNF/G/Ag ternary hydrogel is demonstrated in Fig. 1d. We can clearly observe the existence of C, O and Ag elements. The Ag content is determined as 41.5 wt%, which is consistent with the TGA result.

Fig. S4 displays the antibacterial activity of different groups against E. coli. The blank and CNF/G groups don't show obvious antibacterial activity. Significantly, the CNF/G/Ag composite hydrogels present greatly enhanced antibacterial activity when the Ag content is more than 40 wt%. As shown in Fig. S5, the CNF/G binary composite exhibits the highest cell viability of L02 cells, suggesting the desirable biocompatibility of such designed network, similar to reported results [21]. After integration of Ag NPs into the CNF/G scaffold, the cell viabilities have been largely improved compared to the pure Ag NPs group. Particularly, the CNF/G/Ag-40% displays a cell viability of 109% and higher than that of blank group. For low content of Ag NPs ($\leq$40%), the chemical interactions can promote the homogeneous incorporation of Ag NPs into CNF/G network, thus ensuring the biocompatibility of CNF/G/Ag-40% hydrogel dressing for a long time. However, higher content of Ag NPs can't integrate with CNF/G through effective interactions, the surplus Ag NPs dissociated from CNF/G network will lead to cell cytotoxicity.

In consideration of cell cytotoxicity and antibacterial activity, the composite dressing of CNF/G/Ag-40% was chose to evaluate the in vivo wound healing effect using a full-thickness rat cutaneous wound model. The photographs of the traumas after 1 and 14 days' treatment can be found in Fig. 2a. Comparatively, the rat wound treated with injectable CNF/G/Ag hydrogel grown fresh skin and shown almost recovery after two weeks. As displayed in Fig. 2b, the CNF/G/Ag dressing group presented a wound healing rate of 92.6%, which is higher than control (43.4%) and CNF/G (69.1%) groups indicating its better healing ability due to the synergistic effect. Detailedly, Ag NPs have high anti-infective and antibacterial activities. CNF/G nanofibers show desirable biocompatibility, mechanical property, and chemical modifiability. Accordingly, the rational design of such composite dressing through chemical interactions leads to accelerated wound healing process. The number of white blood cells (WBCs) and neutrophils are of great significance to reflect the immune ability and antibacterial activity. We collected their number in the blood of the above rats, as shown in Fig. 3a, b. The rats received treatment groups exhibited obviously higher levels of both WBCs and neutrophils than the blank group after 3-day surgery. However, after 6 days, the treatment with CNF/G/Ag nanocomposite generated plenty of neutrophils and WBCs by synergistic antibacterial effects, contributing to the promoted immune functions for treated young rats.

4. Conclusion

In summary, we report the preparation of a novel injectable CNF/G/Ag ternary hydrogel for efficient wound therapy. The as-prepared aminated Ag NPs (uniform diameter of 30–60 nm) and nanocomposite hydrogel have been characterized by using TEM, SEM, TGA and XRD techniques. In vitro studies reveal that the injectable CNF/G/Ag composite hydrogel possesses low cell toxicity and excellent antibacterial activity. Significantly, the in vivo evaluation for mice cutaneous wound healing demonstrates an improved wound healing ability with a high wound healing rate of 92.6% after two weeks' treatment with CNF/G/Ag dressing. These results strongly support the potential of such efficient CNF/G/Ag ternary hydrogel for fast cutaneous wound healing in nursing care of infants.

CRediT authorship contribution statement

Li Gou: Resources, Writing - original draft. **Mingli Xiang:** Formal analysis. **Xiaoli Ni:** Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matlet.2020.128340.

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